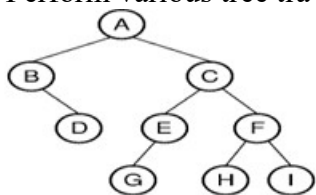


ANALYSIS & DESIGN OF ALGORITHMS LAB

Semester	IV				
Course code					
Category	Professional Core Courses				
Course title	Analysis & Design of Algorithms Lab				
Scheme and Credits	L	T	P	Credits	
	0	0	2	1	
Classwork	50 Marks				
Exam	50 Marks				
Total	100 Marks				
Duration of Exam	02 Hours				

CONTENTS

1. Write a program to implement different sorting techniques.
 - Bubble Sort
 - Insertion Sort
 - Selection Sort
 - Quick Sort
 - Merge Sort
2. Write a program to find minimum cost spanning tree.
3. Write a program to implement travelling sales person problem.
4. Write a program to find Longest Path in a Directed Acyclic Graph.
5. Write a program for Shortest path with exactly k edges in a directed and weighted graph.
6. Write a program find maximum number of edge disjoint paths between two vertices
7. Implement 0/1 Knapsack problem using Dynamic Programming.
8. Perform various tree traversal algorithms for a given tree.



9. Implement N-Queens Problem

COURSE OUTCOMES:

At the end of this course, students will demonstrate the ability to

CO1: Develop and code program for the algorithms and analyze it to determine its computational complexity.

CO2: Identify and analyze worst-case running times of algorithms.

CO3: Model given engineering problem using graph and trees and write the corresponding algorithm to solve the problems.

CO4: Identify and apply the suitable algorithm for the given real-world problem.